

Overlapping Generations in Lean: The Life-Cycle Household

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Abstract

This is the record of a programme to formalise overlapping-generations economies in Lean 4, and of its first stage: the life-cycle household. A household that lives J periods against uninsurable earnings risk and a borrowing constraint solves a finite backward induction, and everything proved for the infinite-horizon Bellman operator with an arbitrary continuation applies stage by stage with no fixed point. Beyond that we prove the finite-horizon sandwich: with k periods still to come, consumption lies between $\kappa_k m$ and $\kappa_k(m + H_k(z))$, where $1/\kappa_k = \sum_{i \leq k} (\mathfrak{p}/R)^i$ is the consumption share of a household with no future income and H_k is the expected present value of the next k incomes; the propensities decrease in the horizon towards the infinite-horizon $\kappa = 1 - \mathfrak{p}/R$. We then add retirement and a pension. With a lump-sum pension the retired phase is an income-fluctuation problem in its own right and everything survives. With a means-tested pension we settle a claim often made informally: a means test does make the policy non-monotone, but not for the reason usually given. Optimal saving rises with cash on hand whatever the continuation, by Topkis, so a value function made non-concave by the test cannot by itself reverse it; what the test reverses is consumption, which jumps down at the wealth where the household gives the pension up, and cash on hand itself, once the taper is steeper than the gross return. Finally we report what the life cycle does for Light's Theorem 1, that saving rises with the interest rate: it holds at every age, exactly, below a critical degree of relative risk aversion of about 3.4 at Aiyagari's calibration, and fails above it, so the life cycle delivers his $\mu = 3$ and not his $\mu = 5$. The programme's target is uniqueness of the stationary equilibrium at relative risk aversion above one, which the infinite horizon could not close: in a life cycle every comparative static is a finite induction, wealth is bounded, and the age-one distribution is common to every rate.

JEL: C63, D52, E21. *Keywords:* overlapping generations, life cycle, Bewley models, formal verification, Lean.

*The Lean development described here is at <https://github.com/LeanEconomics/LeanEconomics>. It was written mostly by Claude (Anthropic), an AI coding model, working under the direction of Robert Kirkby (Victoria University of Wellington); every result reported was checked by the Lean kernel. This is the companion to the write-up on Aiyagari (1994) in the same repository. Comments welcome.

1 Introduction

The companion write-up in this repository proved that the stationary equilibrium of Aiyagari (1994) is unique at his logarithmic calibration, and located exactly what stands in the way for relative risk aversion $\mu \in \{3, 5\}$. Every obstacle it met traces to the fixed point of the infinite-horizon problem: a condition on the response of consumption to the interest rate fails in a tail of wealth that the stationary distribution reaches; an induction that would prove the response bound has a step multiplier $1/\bar{p} > 1$, so it cannot close as a supremum argument; and the only floor on capital supply comes from an ergodic identity that loses a factor $\|u'\|^2/\text{Var}$. Overlapping generations remove all three at once. A household that lives J periods solves a finite backward induction from an explicit terminal condition, so a multiplier $1/\bar{p}$ per step compounds to $\bar{p}^{-J}$, which at Aiyagari's numbers is about 1.03. Wealth is bounded by J periods of accumulation, so any condition needs checking on a known bounded region. And the distribution of the youngest cohort is common to every interest rate, so an ordering of policies pushes forward to an ordering of the whole age profile of distributions by induction on age, with no Doeblin atom and no uniqueness of a stationary distribution.

The target of the programme is uniqueness of the stationary equilibrium of a stochastic life-cycle economy in the manner of Huggett (1996) at $\mu \in \{3, 5\}$. Section 2 records the first stage, the life-cycle household, which is formalised. Section 7 records the rest of the programme, the classical results of Diamond (1965) included, and what each stage needs. As in the companion, every result stated with a formal name has been checked by the Lean kernel against Mathlib; the development is at <https://github.com/LeanEconomics/LeanEconomics>, directory `LeanEconomics/OLG`.

2 The life-cycle household

2.1 Stages

The household of the companion write-up is kept: earnings $y(z)$ follow a finite Markov chain π , the gross return is $R = 1 + r$, cash on hand is $m = Ra + y(z)$, saving $a' \in [0, \bar{a}]$ is bounded below by the borrowing limit and above by an artefactual cap, utility is CRRA with $u'(c) = c^{-\mu}$ and $\beta \in (0, 1)$. What changes is the horizon: the household lives J periods and leaves nothing behind. Count periods from the end. With k periods still to come after today, the continuation value is the k -th Bellman iterate from zero,

$$V_k = T^k 0, \quad (Tv)(a, z) = \max_{a'} \{u(m - a') + \beta \sum_{z'} \pi(z, z') v(a', z')\},$$

and the household with k periods to come saves $g_k = \text{policyOf } V_k$ and consumes $c_k = m - g_k$ (`stageValue`, `stagePolicy`, `stageConsumption`). Age j of a J -period life is stage $k = J - 1 - j$. In the last period the household saves nothing and consumes its cash on hand (`stagePolicy_zero`, `stageConsumption_zero`).

This is the finite-horizon problem with a flat age profile of earnings: the same income

process at every age. An age profile, retirement and survival risk need income to be a parameter of the Bellman step rather than a field of the structure, and are deferred to the next stage of the programme.

2.2 What transfers without a fixed point

The companion development proved its household results for the Bellman operator applied to an arbitrary continuation v , and only then passed to the fixed point. Here there is no fixed point to pass to, and each result is read at $v = V_k$:

- consumption is positive at every stage when marginal utility is unbounded at zero (`stageConsumption_pos`);
- the stage value has concave slices, so the optimal action is unique and both saving and consumption rise with wealth (`stagePolicy_mono`, `stageConsumption_mono`);
- Carroll–Kimball: consumption is concave in wealth at every stage (`concaveOn_stageConsumption`);
- the Euler inequality under the cap, $\beta R \mathbb{E}[u'(c_k(a', z')) \mid z] \leq u'(c_{k+1}(a, z))$, at every state (`stage_euler_le`), and the Euler inequality above the floor, the reverse, at every state where the household saves (`stage_euler_ge`).

2.3 The finite-horizon sandwich

Let $\mathfrak{p} = (\beta R)^{1/\mu}$ and define the finite-horizon minimal propensities to consume by

$$\kappa_0 = 1, \quad \kappa_{k+1} = \frac{\kappa_k R}{\mathfrak{p} + \kappa_k R}, \quad \text{so that} \quad \frac{1}{\kappa_k} = \sum_{i=0}^k \left(\frac{\mathfrak{p}}{R} \right)^i$$

(`stageMPC`): the share of cash on hand consumed by a household with k periods to come and no future income. Let $H_0 = 0$ and $RH_{k+1}(z) = \sum_{z'} \pi(z, z')(y(z') + H_k(z'))$, the expected present value of the next k incomes (`stageHumanWealth`).

Theorem 2.1 (The finite-horizon sandwich). *Assume CRRA utility with marginal utility unbounded at zero, and that the cap is slack at every stage. Then for every k , every z and every $a \in [0, \bar{a}]$,*

$$\kappa_k m \leq c_k(a, z) \leq \kappa_k (m + H_k(z)).$$

Formal statement: `IncomeFluctuation.stageMPC_mul_le_stageConsumption`,
`IncomeFluctuation.stageConsumption_le_stageHumanWealth`.

Each bound is one Euler step, applied k times. The lower bound: if consumption against v is at least κm , the Euler inequality above the floor at a state where the household saves gives $c^{-\mu} \leq \beta R \mathbb{E}[(\kappa R a')^{-\mu}]$ with $a' = m - c$, hence $\kappa R(m - c) \leq \mathfrak{p}c$ and $c \geq \kappa R(\mathfrak{p} + \kappa R)^{-1}m$; at a corner $c = m$ (`crra_stageMPC_step`). The upper bound: if consumption against v is at most $\kappa(m + H(z))$, the Euler inequality under the cap and Jensen for the negative power give

$c^{-\mu} \geq \beta R (\mathbb{E}[c'])^{-\mu}$ with $\mathbb{E}[c'] \leq \kappa R(a' + H'(z))$, hence $c \leq \kappa R(\mathfrak{p} + \kappa R)^{-1}(m + H'(z))$ (`ccra_stageUpper_step`). The recursion for κ is the same on both sides, which is why the sandwich is exact in the homogeneous case.

The propensities decrease in the horizon towards the infinite-horizon $\kappa = 1 - \mathfrak{p}/R$ of the companion write-up: $\kappa \leq \kappa_k$ for every k (`minMPC_le_stageMPC`), so a household with less life left consumes a larger share of its cash, and the infinite-horizon bounds are the limiting case. A first consequence for the programme is an explicit bound on saving, $g_k(a, z) \leq (1 - \kappa_k)(Ra + y_{\max})$ (`stagePolicy_le`), which iterated from zero wealth at the start of life bounds the wealth of every cohort.

3 Retirement, a pension, and the means test

3.1 Retirement with a lump-sum pension

The household works until age J_r and is retired after it, receiving a pension in place of its labour earnings. With a *lump-sum* pension p the retired household faces the same problem as the working one with income replaced by the constant p , so the retired phase is itself an income-fluctuation problem (`withPension`) and everything in Section 2 applies to it unchanged. The life-cycle value is assembled by iterating the working Bellman operator on the retired value: `lifeValue( $n_r, n_w$ )` is the value of a household with n_w working and n_r retired periods still to come (`lifeValue`), so a household of age $j < J_r$ in a life of J periods sits at $n_w = J_r - j$ and $n_r = J - J_r$. Saving and consumption rise with wealth at every age (`lifePolicy_mono`, `lifeConsumption_mono`).

Two facts about the pension itself are worth recording. A retiree's human wealth is the same in every earnings state, because the pension does not depend on the earnings shock (`stageHumanWealth_withPension_const`), and it never exceeds the perpetuity value of the pension, $H_k \leq p/r$ (`stageHumanWealth_withPension_le`). With Theorem 2.1 the retiree's consumption therefore satisfies

$$\kappa_k m \leq c_k(a) \leq \kappa_k \left(m + \frac{p}{r} \right)$$

at every retired age. And cash on hand is strictly increasing in assets (`withPension_resources_strictMono`). Both of these fail once the pension is means-tested.

3.2 Means testing

A pension is means-tested when the amount paid falls with the recipient's assets. We write the schedule as a function $p(\cdot)$ of assets and cash on hand as $m(a) = Ra + p(a)$ (`cash`), and consider three schedules (`lumpSum`, `tapered`, `cliff`): a lump sum; a taper, withdrawn at rate τ per unit of assets above a threshold and never negative; and a cliff, withdrawn entirely above the threshold.

It is often argued that a means test makes the household's policy function non-monotone, and the reason usually given is that the test makes the value function non-concave. The

argument as usually given is wrong, and the conclusion is nevertheless true, for a different reason. Three results separate the two.

Theorem 3.1 (Monotonicity in cash on hand is never lost). *Let period utility be strictly concave on the positive half-line and let W be any continuation value. If $m_1 < m_2$ and x_i is optimal at m_i , then $x_1 \leq x_2$.*

Formal statement: *OLG.isOptimalSaving_mono.*

The proof is Topkis. Adding the two optimality inequalities gives $u(m_1 - x_1) + u(m_2 - x_2) \geq u(m_1 - x_2) + u(m_2 - x_1)$; if $x_2 < x_1$ the two right-hand consumptions are strict convex combinations of the two left-hand ones with the same sum, so strict concavity makes the reverse inequality strict, a contradiction. The continuation enters only through terms that cancel. So a non-concave value function — which a means test certainly produces — cannot by itself make saving fall with cash on hand, whatever the shape of W .

What a non-concave continuation does produce is a jump, and the jump falls on consumption.

Theorem 3.2 (Consumption falls when wealth rises). *There are a continuation value arising from a means-tested pension and two levels of cash on hand, the second larger, at which optimal consumption is strictly smaller. With log utility, a unit gross return, and a pension of one withdrawn entirely above assets of one: at $m = 6$ the household saves exactly 1 and consumes 5; at $m = 8$ it saves 4 and consumes 4.*

Formal statement: *OLG.exists_consumption_decrease.*

At $m = 6$ the household keeps the pension, saving the most it can without losing it; at $m = 8$ the pension is no longer worth keeping, so the household gives it up and saves the proceeds. Saving rises from 1 to 4, as Theorem 3.1 says it must, and since it rises by more than wealth does, consumption falls. Both optima are exact: the objective is log of a quadratic on each side of the threshold, and the comparison is checked by the kernel.

The third result is the genuine non-monotonicity of the saving policy, and it comes from the budget rather than from the value function. A retiree's pension is assessed each period on the assets then held, so the schedule enters cash on hand directly.

Theorem 3.3 (A gentle taper is harmless; a cliff is not). *If $0 \leq \tau < R$ then $a \mapsto Ra + p(a)$ is strictly increasing under the taper (*cash_strictMono_tapered*), and so, by Theorem 3.1, saving is monotone in assets (*isOptimalSaving_mono_assets*). Under a cliff it is not: if $a \leq \bar{a} < b$ and $R(b - a) < p$ then cash on hand is strictly smaller at b (*cash_lt_cash_of_cliff*), and there are two asset levels, the second larger, at which the household saves strictly less: with a concave continuation, at assets 1 it saves $1/2$ and at assets $3/2$ only $1/4$.*

Formal statement: *OLG.exists_saving_decrease_in_assets.*

The continuation in the second half of Theorem 3.3 is $\log(x + 1)$, which is concave: nothing is hidden in it. The non-monotonicity is entirely the budget's, and the threshold is sharp, τ against R . A taper gentler than the gross return leaves the household's saving monotone in assets no matter how badly the test distorts the value function; a taper steeper than the gross return, or a cliff, makes a richer household poorer in the only sense that matters for today's decision.

3.3 Which policy, and in which variable

The two non-monotonicity results are about different objects, and it is worth separating them in the household's own state space, assets a and the Markov income shock z .

Next period's assets are monotone. For each z , the optimal a' is nondecreasing in a whatever the continuation value, with no concavity assumed of it:

Theorem 3.4 (a' rises with a , at each z). *Let $a < a'$ both lie in the asset region. Then for every continuation v and every z , $g_v(a, z) \leq g_v(a', z)$.*

Formal statement: *IncomeFluctuation.policyOf_mono_of_lt.*

This is Theorem 3.1 read in the state space: cash on hand rises with assets, so optimal saving does. The repository already had this statement (`policyOf_mono`) but with concave slices of the continuation as a hypothesis, which is exactly what a means test destroys; the Topkis argument removes the hypothesis, at the cost of requiring the two asset levels to be distinct. The continuation cancels between the two optimality inequalities and only strict concavity of period utility survives.

Consumption is not monotone, and the contrast is sharp rather than an artefact of proof technique. Consumption is $m(a) - g_v(a, z)$, and both terms rise with a ; when the continuation is non-concave the second can rise faster, which is exactly Theorem 3.2. So monotone consumption genuinely requires concavity of the continuation, while monotone saving does not.

Two things follow for the means test. Where the pension enters only next period's problem, so that today's cash on hand is unaffected, Theorem 3.4 applies verbatim: next period's assets still rise with this period's, and only consumption misbehaves. Where the test is assessed on the assets currently held, the hypothesis of Theorem 3.4 is the one that fails: cash on hand is no longer increasing in a , and with it the policy. That is the whole of Theorem 3.3, and the dividing line is τ against R .

3.4 At realistic parameters

The three theorems predict exactly which of the two channels operates for which schedule, and a calibrated retiree confirms it. We solve a deterministic retiree of 25 years by backward induction with global search over a grid of 6001 asset levels — the value function is not concave, so a first-order condition cannot be used — at $r = 4\%$, $\beta = 0.96$, relative risk aversion 3, a full pension of 0.40 of mean earnings and a free area of 3 mean-earnings units (`numerics/meanstest.m`). The taper of 0.078 per unit of assets is Australia's asset test, three dollars a fortnight per thousand dollars.

Pension schedule	Cash on hand	Saving	Consumption
Lump sum	increasing	increasing	increasing
Taper 0.078 (Australia)	increasing	increasing	falls by 0.025–0.046
Taper 0.50	increasing	increasing	falls by 0.13–0.24
Taper 1.20 (above R)	falls	falls by 0.004	falls by 0.55–0.70
Cliff	falls by 0.40	falls by 0.32–0.37	falls by 0.55–0.70

Every entry is what the theorems require. Saving falls with assets in exactly the two rows where cash on hand does, which are exactly the two rows whose taper exceeds the gross return, as Theorem 3.3 says; in the other three it rises, as Theorem 3.1 says it must. Consumption falls in every row with a test of any kind, as Theorem 3.2 says it can, and the drop sits exactly at the free area and grows with age, from 0.025 of mean earnings at the start of retirement to 0.046 twenty years in under the Australian taper. The mechanism is the one the witness isolates: just above the free area the effective return to saving is only $R - \tau$ until the pension is exhausted, so the household jumps its saving toward the far side of the taper range and cuts consumption to pay for it.

The effect is not confined to retirees. A worker in the last year before retirement, facing the Australian taper for the whole of retirement, has a monotone saving policy and a consumption policy that falls by 0.025 of mean earnings at the same threshold: the non-concavity is inherited through the continuation value, and it is again consumption that carries it.

4 Stationary equilibrium

Every cohort is born with no assets and an earnings state drawn from a fixed law ν , lives J periods, and is replaced; cohorts have equal weight. The distribution of the youngest cohort is therefore the same at every interest rate, and the distribution of every older cohort is that law pushed forward by the policies of the ages in between. This is the structural advantage of the life cycle over the infinite horizon, and it removes most of the machinery the stationary case needed.

4.1 The aggregate is a finite composition

Because assets at birth are degenerate at zero, no measure theory is required. Let

$$(\mathcal{S}_k h)(a, z) = \sum_{z'} \pi(z, z') h(g_k(a, z), z')$$

be the expectation of h next period for a household with k periods still to come (`stageStep`). The assets a household holds over the rest of its life, summed across ages and seen from its current state, satisfy

$$A_0(a, z) = a, \quad A_{k+1}(a, z) = a + (\mathcal{S}_{k+1} A_k)(a, z)$$

(`cohortAssets`), and aggregate capital per head is $K = \frac{1}{J} \sum_z \nu(z) A_{J-1}(0, z)$ (`olgCapital`). There is no pushforward measure, no Feller property, no stationary distribution, no Doeblin condition and no uniqueness of an invariant law: the equilibrium object is a finite sum of finite sums, and the recursion above is its definition.

4.2 The chain

Theorem 4.1 (Light’s Theorem 2 for the life cycle). *Suppose two economies share the earnings chain and their policies are ordered at every age, $g_k^{(1)} \leq g_k^{(2)}$ on the asset region. Then $A_k^{(1)} \leq A_k^{(2)}$ for every k , and aggregate capital is ordered, $K^{(1)} \leq K^{(2)}$.*

Formal statement: *IncomeFluctuation.cohortAssets_le, IncomeFluctuation.olgCapital_le_of_stagePolicy_le.*

The proof is an induction on age with two ingredients, and both are already available. The class of functions increasing in assets is preserved by $\mathcal{S}_k$ (`monoRegion_stageStep`), because next period’s assets rise with this period’s at each earnings state and for any continuation, which is Theorem 3.4. And ordered policies keep ordered functions ordered (`stageStep_le_stageStep`): the higher-rate economy evaluates a larger function at a larger asset level. Note what is *not* needed. The infinite-horizon version of this step required monotone transitions, a coupling argument and first-order stochastic dominance of the stationary distributions; here the whole statement is an inequality between two functions on the same finite state space, because the distribution the two economies start from is literally the same.

4.3 A floor on saving from primitives

Existence needs capital supply at the top of the rate interval to exceed demand. In the infinite horizon the only floor available came from an ergodic identity that loses a factor $\|u'\|_\infty^2 / \text{Var}$, with the consequences for the constants reported in the companion write-up. The life cycle needs nothing of the kind: the upper sandwich of Theorem 2.1 bounds consumption at every age, and what is not consumed is saved.

Proposition 4.2 (Affine floor on saving). *At every age and every state in the region,*

$$g_k(a, z) \geq (1 - \kappa_k)(y_{\min} + Ra) - \kappa_k H_k(z).$$

Formal statement: *IncomeFluctuation.le_stagePolicy.*

Iterated from zero assets at birth this is an explicit floor on the assets of every cohort, in the primitives alone: the discount factor, the rate, the earnings bounds and the horizon.

4.4 Where the household can be

Proposition 4.2 inherits from Theorem 2.1 the hypothesis that the artefactual cap is slack, and the convenient way to state that is slack everywhere in $[0, \bar{a}]$. That hypothesis is not innocent, and it fails exactly where the floor is wanted.

The one-step bound gives $g_k(a, z) \leq (1 - \kappa_k)(Ra + y_{\max})$, so $[0, \bar{a}]$ is carried into itself only if $(1 - \kappa_k)(R\bar{a} + y_{\max}) \leq \bar{a}$, which needs $(1 - \kappa_k)R < 1$. As k grows κ_k falls to $1 - \mathfrak{p}/R$, so the requirement is $\mathfrak{p} < 1$, that is $\beta R < 1$: the household must be impatient. Above the rate of time preference no cap is forward invariant at any level, because a patient household sitting at the cap wants to save past it. The cap then binds, the Euler inequality under it is

unavailable, and the slackness hypothesis is false rather than merely unverified. A theorem carrying it says nothing there.

This matters because a stationary life-cycle equilibrium sits above the rate of time preference. In the calibration of Section 4.11 with $\lambda = 1/\beta - 1 = 4.17\%$, capital supply crosses demand near 5%, and the rate at which the *provable* floor crosses demand is near 9%. Both are in the region where no cap is slack everywhere.

A cohort that begins life with nothing never visits the top of $[0, \bar{a}]$. The development therefore carries a family of regions, one per stage, that the stage policies map down through, and asks for slackness only on those.

Definition 4.3 (Reachable family). A reachable family assigns to each stage k a set of asset levels such that every set sits inside $[0, \bar{a}]$, the stage- $(k + 1)$ policy carries the stage- $(k + 1)$ set into the stage- k set, and $g_k < \bar{a}$ on every set.

Formal statement: `IncomeFluctuation.StageRegions`.

The whole interval $[0, \bar{a}]$ is a reachable family exactly when the cap is slack on it (`allRegions`), so nothing is lost: Theorem 2.1 and Proposition 4.2 are the constant-family instances of statements that hold on any reachable family (`stageConsumption_le_stageHumanWealth_on`, `le_stagePolicy_on`).

What a cohort visits is bounded by feasibility alone. Set $\rho_0 = 0$ and $\rho_{j+1} = R\rho_j + y_{\max}$: a household that cannot carry forward more than its cash on hand holds at most ρ_j after j periods of life, whatever the earnings draws.

Proposition 4.4 (A cohort's own regions). *If $\rho_{K+1} < \bar{a}$ then the sets $[0, \rho_{K-k}]$, empty for $k > K$, form a reachable family for a cohort of length $K + 1$ that starts life with nothing.*

Formal statement: `IncomeFluctuation.reachRegions`.

The bound is deliberately crude, using none of the damping that optimal saving supplies, and that is what keeps it free of the sandwich it is used to prove. Its cost is a larger cap. At $r = 10\%$ over sixty periods ρ_{K+1} is about 9,044 where the optimal path reaches about 82. Since the cap is an artefact of the state space rather than an economic object, a cap of ten thousand is as acceptable as a cap of one hundred, and unlike slackness everywhere it is a condition that can actually be met. The sandwich and the floor then hold along a cohort's path with no hypothesis on the cap beyond $\rho_{K+1} < \bar{a}$ (`stageConsumption_le_stageHumanWealth_reach`, `le_stagePolicy_reach`).

At the bottom of the rate interval the same crude bound does the other half of the work on its own. When $R < 1$ the sequence ρ_j converges to $y_{\max}/(1 - R)$, so cohort assets are bounded without any appeal to optimality: at $r = -7\%$ the bound is 42.1 against a capital demand of 56.3.

4.5 Uniqueness

Theorem 4.5 (Single crossing). *A capital supply that never falls with the rate meets a strictly falling demand at most once.*

Formal statement: `IncomeFluctuation.olgEquilibriumRate_unique`.

With Theorem 4.1 this reduces uniqueness of the stationary equilibrium to Light’s Theorem 1, that saving rises with the rate at every age, which is the subject of Section 5: it holds at Aiyagari’s calibration for relative risk aversion up to about 3.4, so the chain closes at his $\mu = 3$ and not at his $\mu = 5$.

4.6 Continuity, and existence

The last input is continuity of capital supply in the rate. In the infinite horizon this was a substantial development, because the stationary distribution itself had to be shown to move continuously. Here there is no stationary distribution: capital supply is a finite composition of finitely many policies, so all that is needed is that each policy moves continuously with the rate, and that continuity composes.

We take the route of the companion write-up and put the interest rate into the state, so that Berge’s theorem gives continuity in assets and the rate jointly rather than continuity in a parameter. That device was used there at the fixed point, but everything it rests on — that the feasible sets and the objectives agree on the slice — holds for an arbitrary continuation. So it works stage by stage, and here it is easier, because the slice of the augmented Bellman iterate is the iterate of the sliced programme by a plain induction, with no fixed point to identify (`sliceAt_bellman`, `sliceAt_augStageValue`). The augmented argmax at each stage is then a singleton, since the sliced one is by Carroll–Kimball concavity of the iterates (`argmax_aug_stage_eq_singleton`), so the stage policy is jointly continuous (`continuousOn_augStagePolicy`) and agrees with the sliced one (`augStagePolicy_eq`). Running the cohort recursion in the augmented state (`augCohortAssets`) gives a composition of continuous maps that reads, at a rate in range, as capital supply at that rate (`augCohortAssets_eq`, `olgSupply_eq`).

Theorem 4.6 (Capital supply is continuous). *$r \mapsto K(r)$ is continuous on $[r_{lo}, r_{hi}]$.*
 Formal statement: *`IncomeFluctuation.continuousOn_olgSupply`.*

Theorem 4.7 (Existence). *If capital supply lies below demand at the bottom of the interval and above it at the top, and demand is continuous, then the two meet: there is a rate in $[r_{lo}, r_{hi}]$ at which the capital the cohorts accumulate equals the capital the firm demands.*
 Formal statement: *`IncomeFluctuation.exists_olgEquilibrium`.*

Both hypotheses of Theorem 4.7 are about the solved model. The rest of this section replaces them with inequalities that never mention the household’s policy or value function.

4.7 The ceiling, from the budget alone

Proposition 4.8 (A forward-invariant level caps supply). *If $0 \leq B \leq \bar{a}$ and $RB + y_{\max} \leq B$, then capital per head is at most B .*
 Formal statement: *`IncomeFluctuation.olgCapital_le_of_invariant`.*

A household starting below B cannot carry forward more than its cash on hand, which is at most $RB + y_{\max} \leq B$, so it never leaves $[0, B]$ and over $K + 1$ periods holds at most

$(K + 1)B$. No optimality, no sandwich, no slackness. Such a B exists whenever $R < 1$, taking $B \geq y_{\max}/(1 - R)$, which is exactly where a Cobb–Douglas demand curve is large.

That is the cheapest ceiling, and it is not the sharpest. The constraint-incidence bound of Theorem 5.7 says the household will not want to carry that much: consuming at least $\kappa_k(m + H_k(z))$ it saves at most $(1 - \kappa_k)(y_z + Ra) - \kappa_k H_k(z)$, and the forward-invariant level that follows is about half the size.

Proposition 4.9 (The sharper ceiling). *If $B \leq \bar{a}$ and, at every age and earnings state, $(1 - \kappa_j)(y_w + RB) - \kappa_j H_j(w) \leq B$, then capital per head is at most B .*

Formal statement: *IncomeFluctuation.olgCapital_le_of_incidence*.

At Aiyagari’s calibration (`numerics/ceiling.m`):

	$r = -7\%$	$r = -6\%$	highest r_{lo} that works
Capital demand	56.25	28.13	
Ceiling from the budget	42.52	49.61	−6.73%
Ceiling from the incidence bound	22.17	25.62	−5.85%

So the margin roughly doubles at a fixed endpoint, and the endpoint can be taken nearly a point closer to zero. The household still enters only through κ and H , both explicit recursions in the primitives, so Theorem 4.12 keeps its character; the price is one extra hypothesis, that the artefactual cap is slack at the low rate, which by Section 4.4 is not vacuous below the rate of time preference (`IncomeFluctuation.exists_olgEquilibrium_of_sharpBounds`).

4.8 The floor, as an affine bound on accumulated assets

Iterating Proposition 4.2 is not a matter of tracking a distribution over assets. An affine lower bound suffices:

$$F_k(z) + G_k a \leq A_k(a, z),$$

with a slope G common to all earnings states and an intercept F carrying one value per state,

$$G_0 = 1, \quad G_{k+1} = 1 + G_k(1 - \kappa_{k+1})R,$$

$$F_0 = 0, \quad F_{k+1}(z) = \sum_{z'} \pi(z, z') F_k(z') + G_k [(1 - \kappa_{k+1})y_z - \kappa_{k+1} H_{k+1}(z)].$$

Proposition 4.10 (Affine floor on accumulated assets). *On any reachable family, $F_k(z) + G_k a \leq A_k(a, z)$; hence capital per head is at least $\sum_z \nu_z F_K(z)/(K + 1)$.*

Formal statement: *IncomeFluctuation.cohortFloor_add_mul_le_cohortAssets*.

, *IncomeFluctuation.le_olgCapital*

The state dependence of F is the whole content of the bound, not a refinement. Reading Proposition 4.2 at the worst earnings state instead, which is the only way to get a bound uniform in z , gives a floor that is identically zero at every rate in the calibration of Section 4.11, while the state-dependent one reaches 4.75 at $r = 10\%$ against a demand of 3.13.

Averaging against earnings weights that the chain leaves alone collapses the vector recursion to a scalar one, and then the sign of each increment has an exact answer.

Theorem 4.11 (The floor rises exactly when the household is patient). *Let $\bar{y}$ and $\bar{H}_k$ be the averages of earnings and of human wealth under weights ν satisfying $\nu\pi = \nu$. If $\bar{p} \geq 1$, equivalently $\beta R \geq 1$, then $\kappa_k \bar{H}_k \leq (1 - \kappa_k) \bar{y}$ at every age, so the aggregate floor never falls and is never negative.*

Formal statement: *IncomeFluctuation.stageMPC_mul_sum_stageHumanWealth_le.*
, IncomeFluctuation.sum_mul_cohortFloor_mono

The increment of the aggregate floor at each age is $G_k \kappa_{k+1} \bar{y}$ times the slack in that inequality, and the inequality reverses below $\bar{p} = 1$. So the provable floor is flat under the rate of time preference and climbs above it, which is what Section 4.11 shows and why the upper endpoint of the rate interval has to be taken above $1/\beta - 1$. The proof is one induction: writing $v_k = (1 - \kappa_k)/\kappa_k$ and $u_k = \bar{H}_k/\bar{y}$, both satisfy first-order recursions, $v_{k+1} = (\bar{p}/R)(1 + v_k)$ and $u_{k+1} = (1 + u_k)/R$, from the same starting point, and $\bar{p} \geq 1$ keeps the first above the second.

4.9 Existence, with both ends discharged

Theorem 4.12 (Existence from primitives). *Fix a rate interval and continuous demand D . Suppose there is a B with $0 \leq B \leq \bar{a}$, $(1 + r_{lo})B + y_{\max} \leq B$ and $B \leq D(r_{lo})$; and suppose $\rho_{K+1} < \bar{a}$ and $D(r_{hi}) \leq \sum_z \nu_z F_K(z)/(K+1)$ at r_{hi} . Then there is a rate in the interval at which the capital the cohorts accumulate equals the capital the firm demands.*

Formal statement: *IncomeFluctuation.exists_olgEquilibrium_rate.*

Nothing in those hypotheses refers to the household's decisions. They are inequalities among the discount factor, the interest rate, the earnings process, the horizon and the demand curve, with ρ , F and G explicit finite recursions in those primitives. With Theorem 4.1 and Theorem 4.5 the life-cycle economy therefore has the same three-part structure as the stationary one — existence, monotone supply, single crossing — with every piece proved and only Light's Theorem 1 assumed.

4.10 The propensity in closed form, and the floor with it

The last hypothesis can be put in closed form too, and doing so turns on knowing not just that κ_k exceeds its infinite-horizon limit but how fast it settles.

Proposition 4.13 (The finite-horizon propensity). *Write $q = \bar{p}/R$. Then $\kappa_k \sum_{i \leq k} q^i = 1$, and hence $\kappa_k(1 - q^{k+1}) = 1 - q$: the propensity exceeds $1 - q$ by exactly $\kappa_k q^{k+1}$.*

Formal statement: *IncomeFluctuation.stageMPC_mul_mpcSum.*

, IncomeFluctuation.stageMPC_mul_one_sub_pow

The defining recursion $\kappa_{k+1} = \kappa_k R/(\bar{p} + \kappa_k R)$ is exactly $1/\kappa_{k+1} = 1 + q/\kappa_k$, so the reciprocal accumulates the geometric series and nothing else. The same sum tames the slope. Writing $S_k = \sum_{i \leq k} q^i$, the recursion $G_{k+1} = 1 + G_k(1 - \kappa_{k+1})R$ has an age-dependent coefficient, but $W_k = G_k S_k$ obeys $W_{k+1} = S_{k+1} + \bar{p}W_k$, which does not. Then $S_{k+1} \geq 1$ and $(1 - q)S_k \leq 1$ give the bound that matters.

Proposition 4.14 (A slope that grows). $G_k \geq (1 - q) \sum_{j \leq k} b^j$.

Formal statement: *IncomeFluctuation.mul_geomSum_le_cohortSlope*.

Average human wealth is a geometric sum as well, $\bar{H}_{k+1} = \bar{y}R^{-1} \sum_{i \leq k} R^{-i}$, so every factor in the increment of Theorem 4.11 is explicit and the aggregate floor becomes a closed-form expression in the discount factor, the interest rate, average earnings and the horizon, with no recursion in the model left to run (*IncomeFluctuation.geomFloor_le_sum_mul_cohortFloor*).

Theorem 4.15 (Existence with both ends in closed form). *Under the conditions of Theorem 4.12, with the floor hypothesis replaced by the closed-form expression and with $1 \leq b \leq R$ at r_{hi} , a stationary equilibrium exists.*

Formal statement: *IncomeFluctuation.exists_olgEquilibrium_of_geomBounds*.

The two conditions on b say the household is patient enough for the floor to climb and impatient enough for its propensity to settle. Knowing the rate is worth a great deal. Using only $\kappa_k \geq 1 - q$ and $G_k \geq 1$, the resulting bound first exceeds Cobb–Douglas demand at $r \approx 610\%$, where a lifetime of feasible saving reaches 6×10^{50} ; using Proposition 4.14 it does so at $r \approx 18\%$, where the same quantity is 3.8×10^5 (*numerics/closedform.m*, *numerics/bisectC.m*). What is still given away is the exact propensity inside the increment and the exact slope: at $r_{hi} = 10\%$ the closed form gives 0.48 against a true floor of 4.75 and a demand of 3.13, so at a calibrated rate it is the finite recursion of Section 4.11 that has to be evaluated, not the closed form.

4.11 What the numbers say

The calibration is the one used throughout: $J = 60$, $\beta = 0.96$, relative risk aversion 3, the Tauchen discretisation of Aiyagari’s earnings process with $\rho = 0.9$ and $\sigma = 0.4$, and a Cobb–Douglas technology with $\alpha = 9/25$ and $\delta = 2/25$, so capital demand is $\alpha/((1 - \alpha)(r + \delta))$. Capital supply is the average asset holding across the sixty cohorts, each born with nothing. The last column is not the solved supply but the floor that Proposition 4.2 provably delivers, iterated across ages with the expectation of H_k taken under the ergodic distribution and Jensen’s inequality applied to the outer maximum (*numerics/olgexist.m*).

r	−4%	0%	$\lambda = 4.17\%$	6%	8%	10%	15%
Capital demand	14.06	7.03	4.62	4.02	3.52	3.13	2.45
Capital supply	1.03	1.98	3.78	4.85	6.18	7.69	12.49
Provable floor	0.00	0.00	0.00	1.52	3.13	4.75	9.40

Supply crosses demand between 4.17% and 6%, so the equilibrium rate exceeds the rate of time preference, which is what Section 4.4 has to accommodate. The provable floor crosses between 8% and 10%, so $r_{hi} = 10\%$ suffices for the upper inequality of Theorem 4.7 and $r_{hi} = 8\%$ does not. The optimal path reaches assets of 82 at $r = 10\%$, against the crude reachable bound of 9,044 and a demand of 3.13; the constants are ordinary, unlike the stationary case, where the corresponding instantiation put r_{hi} within 10^{-27} of λ .

4.12 The theorem checked, hypothesis by hypothesis

Theorem 4.12 and its sharper sibling are only worth as much as the hypotheses they carry, so it is worth checking every one at a calibration. Take the calibration of Section 4.11 with $r_{lo} = -7\%$, $r_{hi} = 10\%$, artefactual cap $\bar{a} = 10^4$ and $B = 30$. Every hypothesis holds, with the margin shown (`numerics/sharpcheck.m`).

Hypothesis	What it asks	Margin
$0 < 1 + r_{lo}$, $0 < 1 + r_{hi}$, $r_{lo} \leq r_{hi}$	the rates are admissible and ordered	0.93, 1.10, 0.17
$r_{lo} + \delta > 0$	demand is continuous on the interval	0.010
$\nu \geq 0$, $\sum \nu = 1$	the earnings weights are a distribution	0.014, exact
cap slack at r_{lo}	$\bar{a} > y_{\max} + R_{lo}\bar{a}$	697
$0 \leq B \leq \bar{a}$	the ceiling level is admissible	30, 9970
forward invariance	$(1 - \kappa_j)(y_w + R_{lo}B) - \kappa_j H_j(w) \leq B$	0.585
$B \leq D(r_{lo})$	the ceiling is below demand	26.25
$\rho_{K+1} < \bar{a}$	a lifetime of feasible saving fits	967
$D(r_{hi}) \leq \sum_z \nu_z F_K(z)/(K+1)$	the floor is above demand	1.628

The constants behind them are: a reachable bound $\rho_{K+1} = 9,033$, a ceiling fixed point 29.42 against a demand of 56.25 at the bottom, and a provable floor of 4.75 against a demand of 3.13 at the top. The equilibrium the theorem then asserts is real: solving the model, capital supply crosses demand at $r = 5.09\%$, comfortably inside the interval.

4.13 How far the hypotheses reach

Sweeping the endpoints (`numerics/sharpsweep.m`) shows where each side gives way.

r_{lo}	-7.9%	-7.0%	-6.5%	-6.0%	-5.5%
Capital demand	562.50	56.25	37.50	28.13	22.50
Ceiling	19.54	22.17	23.81	25.62	27.59
Room for B	542.96	34.09	13.69	2.51	none
r_{hi}	6%	8%	9%	10%	15%
Capital demand	4.018	3.516	3.309	3.125	2.446
Provable floor	1.524	3.129	3.934	4.753	9.397
Reachable bound	1.6×10^3	3.7×10^3	5.8×10^3	9.0×10^3	8.7×10^4

The hypotheses hold for any r_{lo} at or below -5.85% and any r_{hi} at or above 8.38% , so the *tightest* bracket they give is $[-5.85\%, 8.38\%]$, needing a cap of 4.4×10^3 . The true equilibrium at 5.09% sits inside it with room on both sides. With the budget ceiling in place of the incidence one the low endpoint would have to be -6.73% or below, which is the whole of what Proposition 4.9 buys.

Two features of the calibration decide the endpoints, and they are different. At the bottom it is the demand curve's pole at $r = -\delta$ that makes demand large, and the ceiling grows slowly enough to stay under it. At the top it is the patience criterion of Theorem 4.11: below the rate of time preference the provable floor does not move at all, so the upper endpoint cannot be anywhere near $\lambda = 4.17\%$, and it in fact has to be about twice that.

4.14 The ceiling read as an average, not a worst case

Proposition 4.9 bounds every age's assets by one forward-invariant level, and so pays $(K+1)B$ for a cohort that holds nothing when it is born, nothing when it dies, and anything like B only in between. The floor never made that mistake: it carries the affine bound $F_k(z) + G_k a$ and aggregates once at the end. The ceiling can be built the same way, from the same slope G , with the saving ceiling of the incidence bound in place of the saving floor:

$$\bar{F}_0 = 0, \quad \bar{F}_{k+1}(z) = \sum_{z'} \pi(z, z') \bar{F}_k(z') + G_k [(1 - \kappa_{k+1})y_z - \kappa_{k+1}H_{k+1}(z)],$$

with H here the capped, risk-adjusted human wealth of Theorem 5.7.

Proposition 4.16 (The affine ceiling). $A_k(a, z) \leq \bar{F}_k(z) + G_k a$, and hence capital per head is at most $\sum_z \nu_z \bar{F}_K(z)/(K+1)$.

Formal statement: `IncomeFluctuation.cohortAssets_le_cohortCeil.`

`, IncomeFluctuation.olgCapital_le_cohortCeil`

Replacing a worst case over ages by an average over them is worth a factor of fourteen (`numerics/affceil.m`).

	$r = -7\%$	$r = -4\%$	$r = 0\%$	$r = \lambda$	$r = 5\%$
Capital demand	56.25	14.06	7.03	4.62	4.33
Invariant level	22.17	35.11	—	—	—
Affine ceiling	1.54	2.28	3.86	5.98	6.47
True capital supply	0.64	1.03	1.98	3.78	4.25

The affine ceiling sits within a factor of 1.5 to 2.4 of the truth where the invariant level was out by a factor of twenty-six, and it keeps working above the rate of time preference, where no forward-invariant level exists at all. The bottom of the interval then rises from -5.85% to $+2.78\%$, against an absolute limit of 5.09% , which is where capital supply meets demand and where no upper bound on supply can help.

Theorem 4.17 (Existence, both ends read the same way). *With the ceiling at r_{lo} and the floor at r_{hi} both taken as weighted averages of affine bounds on a cohort's accumulated assets, a stationary equilibrium exists.*

Formal statement: `IncomeFluctuation.exists_olgEquilibrium_affine.`

No forward-invariant level and no auxiliary B survive in that statement: the two boundary conditions are the same object built twice, once from the saving ceiling and once from the saving floor, sharing a slope. Each half asks for what it can have. At the bottom the artefactual cap is slack, which holds below the rate of time preference; at the top it cannot be, and the reachable family of Section 4.4 stands in. Checking every hypothesis at $r_{lo} = 2\%$, $r_{hi} = 10\%$ and a cap of 10^4 (`numerics/affcheck.m`), all hold, with the ceiling at 4.81 against a demand of 5.63 and the floor at 4.75 against a demand of 3.13. The bracket $[2\%, 10\%]$ is half the width of the one Section 4.12 could offer, and it still contains the true equilibrium.

4.15 Why the top will not widen

The upper endpoint of 8.38% is set by the provable floor meeting demand, and it is twice the rate of time preference. It is worth knowing whether that is a weakness of the bound or of the economics, and the answer is the first, with a definite cause.

Start with where the floor loses. Running the floor's own recursion on the cohort's *true* mean consumption path reproduces aggregate capital exactly, 6.180 against 6.180 at $r = 8\%$ (`numerics/diag.m`). So none of the loss is in the aggregation, the affine bound or the earnings weights: all of it is in the one inequality $c_k \leq \kappa_k(m + H_k(z))$.

That inequality is tight at the end of life and loose at the start.

Age	0	10	30	59
Consumption, truth over bound	0.81	0.88	0.93	1.00
Assets, floor over truth	—	0.50	0.51	0.48

The leverage is what matters. At birth the household earns 1 and consumes 0.698, so it saves 0.302; the bound allows it to consume 0.858, so the provable saving is 0.142. Overstating consumption by a fifth halves the saving, and the asset floor sits at about half the truth at every age thereafter.

Three ways of tightening that inequality were tried and none works.

Truncating the horizon. Feeding the trivial bound $c \leq m$ into the same Euler step j times rather than k gives $c_k \leq \kappa_j(m + H_j(z))$ for every $j \leq k$, so the minimum over j is available, and the check confirms every branch is valid. It buys nothing: $\kappa_j(m + H_j)$ is decreasing in j , so the minimum is always the full-horizon branch already in use (`numerics/trunc.m`).

Keeping the power mean. The Euler step produces the power mean with exponent $-\mu$, and the proof relaxes it to the arithmetic mean, which is 20% larger here. Substituting the power mean back is not legitimate: the step shifts by gross saving, and a power mean with negative exponent is superadditive, so the shift comes out larger inside than outside, which is the wrong way for an upper bound. The arithmetic version is what makes the shift exact. Carried through anyway the result is invalid, violating the true consumption function by 0.30 and producing a floor of 7.45 at $r = 8\%$ against a true capital supply of 6.18 (`numerics/riskupper.m`).

Keeping the power mean with assets inside. Putting the shift inside the mean avoids superadditivity, at the cost of evaluating tomorrow's bound at a saving *ceiling* rather than at the true saving, which decouples the budget identity from the Euler step. That loses more than the power mean gains: the bound at birth rises from 0.858 to 0.956 against a truth of 0.698 (`numerics/gamma.m`).

So the arithmetic relaxation is not an avoidable slip. Solving the budget identity simultaneously with the Euler step is what produces the sharp $\kappa_{k+1}(m + H_{k+1})$ form, and that is exactly what forces the arithmetic mean.

Finally, the size of the prize. No lower bound on capital supply can exceed the truth, and the true crossing is at 5.09%, so the best any sharpening could do is move the upper endpoint from 8.38% to 5.09%. The whole of that 3.3 points is the young household's consumption

being overstated by a fifth, and that overstatement is a risk effect: the household consumes less than permanent income because earnings are risky and the limit may bind. Capturing it needs the power mean, and the power mean is what the step cannot carry.

4.16 The floor read at the assets held

Section 4.14 fixed the ceiling by refusing to worst-case over ages. The floor has a fault of the same kind, in a different place. Its intercept $F_k(z)$ goes badly negative: at the calibration, 160 of 413 entries are below zero and the worst is -62 , with F_K running from -58.8 at the lowest earnings to $+685$ at the highest (`numerics/floordiag.m`). A cohort's accumulated assets cannot be negative, so all of that is slack the aggregate inherits.

An affine form cannot express the clamp. Reading the floor as a function of assets can:

$$\Lambda_0(z, a) = a, \quad \Lambda_{k+1}(z, a) = a + \sum_{z'} \pi(z, z') \Lambda_k(z', \Phi_{k+1}(z, a)),$$

with Φ the saving floor of Section 5.13, clamped at zero. The proof needs only that Λ_k is increasing in assets, which the clamp preserves.

Proposition 4.18 (The clamped floor). $\Lambda_k(z, a) \leq A_k(a, z)$ on any reachable family, so capital per head is at least $\sum_z \nu_z \Lambda_K(z, 0)/(K+1)$.

Formal statement: `IncomeFluctuation.cohortFloorAt_le_cohortAssets.`

, `IncomeFluctuation.le_olgCapital_at`

	$r = 5\%$	$r = 6\%$	$r = 7\%$	$r = 8\%$	$r = 10\%$
Capital demand	4.327	4.018	3.750	3.516	3.125
Affine floor	0.704	1.524	2.329	3.129	4.753
Clamped floor	2.091	2.588	3.169	3.799	5.187
True capital supply	4.248	4.848	5.492	6.180	7.691

The top of the interval falls from 8.38% to 7.67%. Pairing it with the affine ceiling, whose intercept never goes negative because it is built from the saving ceiling, gives the tightest statement the development offers.

Theorem 4.19 (Existence, both ends read at the assets held). *With the affine ceiling at r_{lo} and the clamped floor at r_{hi} , a stationary equilibrium exists.*

Formal statement: `IncomeFluctuation.exists_olgEquilibrium_clamped.`

Checking every hypothesis at $r_{lo} = 2\%$, $r_{hi} = 8\%$ and a cap of 10^4 (`numerics/clampcheck.m`), all hold, with the ceiling at 4.81 against a demand of 5.63 and the floor at 3.80 against a demand of 3.52. The bracket has come down from the 17 points of Section 4.12 to 6, and the tightest these bounds can give is $[2.78\%, 7.67\%]$, against an absolute limit of 5.09% on both sides.

The same diagnostic closes the other candidate for the top for good. Re-running the power-mean recursion of Section 4.15 with the *true* saving reproduces the truth exactly, so all of its loss is the saving ceiling it has to use instead; and that ceiling is only 1.19 times

the truth at birth, yet still produces a worse bound than the arithmetic one, 0.956 against 0.858. A fifth of an overstatement compounded across fifty-nine steps is what does it. The arithmetic relaxation stays the best available.

5 Theorem 1 in the life cycle: how far it goes

The programme of Section 7 rests on Light’s Theorem 1, that saving rises with the interest rate at every state. In the infinite horizon this is open above logarithmic utility. The life cycle was the reason for hope: a finite backward induction from an explicit terminal condition, with the per-step loss compounding only J times instead of to a fixed point. This section reports what the numbers say, at $\beta = 0.96$, a seven-state Tauchen chain with $\rho = 0.9$ and $\sigma = 0.4$, a zero borrowing limit, $J = 60$, and a flat earnings profile (`numerics/lc_check.m`, `lc_robust.m`, `lc_induct.m`, `lc_verify.m`, `lc_bound.m`, `lc_crit.m`).

5.1 The statement is its own terminal condition

The budget constraint makes two statements identical. Since $a' = Ra + y(z) - c$ at either rate,

$$g_k(a, z; r_2) \geq g_k(a, z; r_1) \iff c_k(a, z; r_2) - c_k(a, z; r_1) \leq (r_2 - r_1)a,$$

so Theorem 1 *is* the affine bound on the rate response of consumption with slope one and zero intercept. That bound holds with equality in the last period of life, where $c_0 = y(z) + Ra$. The question is therefore whether the terminal condition propagates backward, and it is a question about one inequality rather than about a pair of constants to be chosen.

5.2 It propagates, up to a critical risk aversion

At $\mu = 1$ and $\mu = 3$ it propagates exactly. Over all sixty stages and the whole wealth region reachable from zero wealth at birth, the largest violation of $g_k^{(1)} \leq g_k^{(2)}$ is zero, and the consumption bound holds to five parts in 10^{14} ; refining the asset grid from 2000 to 16000 points changes nothing. At $\mu = 5$ it fails. The failure is a genuine feature of the household’s problem, not of the solution method: the violation is identical at every grid size, and endogenous-grid and value-function iteration with a pure grid search agree on it to the resolution of the grid. Saving at zero wealth in the highest income state, in the first period of a sixty-period life, falls as the rate rises:

r	2%	3%	4%	5%	6%
$g(0, z_{\max})$, endogenous grid	1.7783	1.7638	1.7452	1.7234	1.6993
$g(0, z_{\max})$, value iteration	1.7806	1.7606	1.7406	1.7206	1.7006

Between the two lies a critical degree of risk aversion, and it is sharp. Bisecting on μ for the largest violation over all stages gives:

Horizon J	$r = 2\%$	$r = 4\%$	$r = \lambda = 4.17\%$
20	3.79	3.53	3.51
40	3.77	3.39	3.36
60	3.77	3.39	3.36

Theorem 1 holds throughout the life cycle below the critical value and fails above it. The value is stable in the horizon beyond about forty periods, and falls as the rate rises towards the rate of time preference. Aiyagari's $\mu = 3$ is inside it with little room, at $\mu = 3.39$ against 3; his $\mu = 5$ is well outside. So the life cycle buys one of the two open cases and not the other, and it buys $\mu = 3$ only at rates below the rate of time preference, which is where the equilibrium lies.

5.3 Why the natural induction does not close

The backward induction that the transfer lemma suggests compares the two Euler equations at a state where the lower-rate household saves. Writing $b = g_{k+1}^{(1)}(a, z)$ for the pivot, the induction hypothesis and monotonicity of c_k give $c_k^{(2)}(b') \leq c_k^{(1)}(b) + \Delta r b$ at the relevant points, and the step closes provided

$$R_2 \mathbb{E}[(c_k^{(1)}(b) + \Delta r b)^{-\mu}] \geq R_1 \mathbb{E}[c_k^{(1)}(b)^{-\mu}] \left(1 + \frac{\Delta r a}{c_{k+1}^{(1)}(a, z)}\right)^{-\mu}.$$

To first order this asks $\mu(b/c_k^{(1)}(b) - a/c_{k+1}^{(1)}(a, z)) \leq 1/R$: the ratio of assets to consumption may rise from one age to the next by at most $1/(\mu R)$. It does not. At zero wealth in the highest income state the left-hand side is large, because such a household saves a great deal against a certain fall in income, while the right-hand side is about one. The margin is negative from the third stage of life onward for every degree of risk aversion we tried, logarithmic utility included, at -0.12 for $\mu = 1$, -2.86 for $\mu = 3$ and -6.29 for $\mu = 5$ per unit of the rate gap. Keeping the term the step discards, the marginal propensity to consume times the size of the hypothetical violation, bounds the violation by a multiple of Δr rather than eliminating it, which is not enough: the aggregate difference the equilibrium argument must sign is itself of order Δr .

5.4 The reduction, formalised, and its verdict

The backward induction above is proved in Lean, in the form the life cycle makes natural. Because the budget constraint makes the two statements identical (`stagePolicy_le_iff_consumption_le`), the terminal stage needs no hypothesis at all: in the last period both households consume their cash on hand and save nothing (`stagePolicy_mono_zero`). Corners need none either: if the lower-rate household saves nothing there is nothing to violate. What the step needs is one inequality per age and state, which we call the *age-indexed slack condition*: writing b for the lower-rate household's saving at that state and $C'(z')$ for its

consumption next period,

$$R_1 \sum_{z'} \pi(z, z') C'(z')^{-\mu} \leq R_2 \sum_{z'} \pi(z, z') (C'(z') + \Delta r b)^{-\mu} \left(1 + \frac{\Delta r a}{c}\right)^\mu \quad (1)$$

(StageSlack).

Theorem 5.1 (The reduction). *If (1) holds at every age and every state in the region, then saving rises with the rate at every age, and aggregate capital rises with the rate.*

Formal statement: *IncomeFluctuation.stagePolicy_mono_step,*

IncomeFluctuation.stagePolicy_mono_of_stageSlack, IncomeFluctuation.olgCapital_le_of_stageSlack.

With Theorem 4.5 this closes the chain: (1) at every age gives a unique equilibrium rate. The question is therefore whether (1) is true, and the answer is the one reported above, now stated sharply. Condition (1) is exactly the margin tabulated in the previous subsection. At $\mu = 3$ it holds for the last three or four periods of life, with margins +0.68, +0.42 and +0.18 per unit of the rate gap, and fails from about the fifth-to-last, at -0.23 , deteriorating to -2.86 at birth. At $\mu = 1$ it holds for most of the life and still fails for the youngest ages, at -0.12 at birth.

That last observation is the important one. The condition fails even at logarithmic utility, where Theorem 1 is certainly true. So the failure is not a symptom of high risk aversion: the pointwise Euler comparison is simply the wrong instrument in a life cycle, and no sharpening of it that we could find repairs the young ages. Keeping the term the step discards, the marginal propensity to consume times the size of the hypothetical violation, only bounds the violation by a multiple of Δr , and both refinements we tried — an upper bound on the propensity to consume from concavity and the sandwich, and a self-improving bootstrap on the size of the violation — converge to a positive bound rather than to zero.

So $\mu = 3$ is not delivered by this route, and $\mu = 1$ is not either, although the companion write-up proves the latter by the rescaling argument, which is an induction on Bellman iterates and so carries over to the life cycle unchanged. What the reduction does deliver is the end of life, and the exact inequality that any proof must beat.

So the state of the programme is this. Theorem 1 is true in the life cycle for $\mu = 3$, and the statement to prove has the cleanest possible form, its own terminal condition propagated; the reduction to one inequality per age is proved; and that inequality is false for the young, at every degree of risk aversion. What is missing is an argument at zero wealth that does not go through a pointwise comparison of the two Euler equations. The same zero-wealth state is where the infinite-horizon slack condition was tightest, which suggests the two problems share a single missing ingredient rather than two.

5.5 It is not precautionary saving

A household at zero wealth in the highest earnings state saves heavily, and the obvious diagnosis is precautionary: it is insuring against the fall in earnings it expects. If that were the mechanism, precautionary-saving theory would be the place to look for the missing

argument. It is not the mechanism. The same reversal happens in the *certainty-equivalent* household, which faces the conditional mean earnings path and no risk at all, and there everything is in closed form (`numerics/precaut2.m`, `durcheck.m`).

d(saving)/dr at zero wealth	$\mu = 1$	$\mu = 3$	$\mu = 5$	$\mu = 8$
Certainty equivalent (mean path, no risk)	+21.3	+2.81	-0.85	-2.91
With risk	+17.1	+0.92	-2.05	-3.43

The sign flips between $\mu = 3$ and $\mu = 5$ in both columns. Bisecting, the certainty-equivalent threshold is $\mu = 4.33$ against $\mu = 3.39$ with risk: risk moves the threshold by about one unit of risk aversion and does not create the effect.

Without risk the mechanism is exact. A household with J periods, no borrowing constraint and CRRA utility has $c_{t+1} = (\beta R)^{1/\mu} c_t$ and a lifetime budget, so

$$c_0 = \frac{W(R)}{D(R)}, \quad W(R) = \sum_t y_t R^{-t}, \quad D(R) = \sum_t \beta^{t/\mu} R^{t/\mu - t}$$

(`consumption_mul_pathPrice`), and saving at zero wealth rises with the rate exactly when $W(R_2)D(R_1) \leq W(R_1)D(R_2)$ (`consumption_antitone_iff`). Both terms fall with the rate, W because future earnings are discounted harder and D because the optimal path gets cheaper, and the comparison is between the two elasticities. Differentiating,

$$\frac{d \log c_0}{d \log R} = -T_y + \left(1 - \frac{1}{\mu}\right) T_c,$$

with T_y and T_c the Macaulay durations of the two streams, so c_0 falls, and saving rises, exactly when $T_y > (1 - 1/\mu)T_c$. The predicted elasticity matches the exact one to three decimals at every μ we checked. At Aiyagari's numbers $T_y = 14.31$ and $T_c \approx 18.6$, nearly independent of μ , so the condition is $1 - 1/\mu < 0.769$, that is $\mu < 4.33$.

The economics is then plain, and has nothing to do with risk. A higher interest rate discounts future earnings more heavily, which makes the household poorer and pushes consumption down; it also makes a rising consumption path cheaper, which pushes consumption today down only to the extent that the household is willing to tilt its path. A more risk averse household will not tilt, so the second force weakens as μ grows while the first does not, and past $\mu \approx 4$ the household responds to a higher rate by consuming more today. The income stream simply has to be longer than the consumption weights, discounted by the strength of the tilt.

Theorem 5.2 (No risk, and risk aversion at most one). *If $\mu \leq 1$ the comparison holds term by term, whatever the earnings profile: the certainty-equivalent household always saves more at a higher rate.*

Formal statement: *CertaintyEquivalent.humanWealth_mul_pathPrice_le_of_le_one*,
CertaintyEquivalent.consumption_antitone_of_le_one.

For $\mu > 1$ it is the duration comparison, which is a finite computation in the primitives. So a precautionary argument near zero wealth would be aimed at the wrong target: what

has to be controlled is the elasticity of human wealth against the elasticity of the price of the consumption path, and risk enters only as a correction to it.

5.6 The criterion in elasticity units, and the size of the risk correction

The certainty-equivalent criterion is cleanest when written in the intertemporal elasticity of substitution rather than in risk aversion. Since it is linear in $1/\mu$,

$$T_y > \left(1 - \frac{1}{\mu}\right)T_c \quad \Longleftrightarrow \quad \frac{1}{\mu} > 1 - \frac{T_y}{T_c},$$

so the elasticity of substitution must exceed one minus the ratio of the two durations. At Aiyagari's numbers that right-hand side is 0.2306, giving $\mu < 4.33$ as before, and the identity holds to four decimals against the bisected threshold.

With risk the threshold moves down, and the move is what has to be bounded. Write

$$\Delta := \frac{1}{\mu^{\text{risky}}} - \frac{1}{\mu^{\text{CE}}}$$

for the gap in elasticity units and $s = \sigma\sqrt{1 - \rho^2}$ for the conditional standard deviation of log earnings, the dispersion the household actually faces one period ahead. Measuring both thresholds at the same state over fifteen calibrations, $\rho \in \{0, 0.6, 0.9\}$ and σ from 0.025 to 0.4 (`numerics/gapscale.m`, `gapfit.m`):

$\rho = 0.9, \sigma$	0.025	0.05	0.1	0.2	0.4	fitted exponent
Δ	0.0034	0.0070	0.0144	0.0300	0.0612	
Δ/s	0.315	0.320	0.330	0.344	0.351	1.04

The gap is first order in the conditional standard deviation, not second order: fitting $\Delta \sim s^b$ gives $b = 1.04$ at $\rho = 0.9$, 0.89 at $\rho = 0.6$ and 0.93 at $\rho = 0$, and the ratio Δ/s stays between 0.196 and 0.351 across every calibration we ran. That first-order behaviour is itself informative: a precautionary effect on saving would be second order in the dispersion, whereas the probability that the borrowing constraint binds in a low-earnings state next period moves at first order, and that is our reading of where the correction comes from.

Taking the largest ratio observed as the bound, $\Delta \leq 0.36 s$, gives a criterion entirely in the primitives:

$$\frac{1}{\mu} \geq 1 - \frac{T_y}{T_c} + 0.36 \sigma \sqrt{1 - \rho^2}. \quad (2)$$

At Aiyagari's calibration the right-hand side is $0.2306 + 0.0628 = 0.2934$, that is $\mu \leq 3.41$, against the 3.39 that bisection on the full life-cycle model gives. The bound therefore reproduces the measured threshold to within two per cent, and it says the same thing the earlier sections said: the calibration he used for $\mu = 3$ is inside, and $\mu = 5$ is not.

Two caveats. The first inequality of (2) is a theorem about the certainty-equivalent household (`consumption_antitone_iff`, and Theorem 5.2 for $\mu \leq 1$); the constant 0.36 is measured over fifteen calibrations and is not proved. And both are statements at zero wealth

in the highest earnings state, which is where the life-cycle model binds; the reduction of Theorem 5.1 is what would carry them to the rest of the state space, and its own condition fails, so this is a sharp description of where the truth lies rather than a proof of it.

5.7 The channel is the borrowing constraint

The first-order scaling points at the constraint rather than at precaution, and the point can be tested directly: loosen the borrowing limit and watch the threshold. If the constraint is the channel, the risky threshold should walk back to the certainty-equivalent one as the limit approaches the natural one, $y_{\min}/r$. At Aiyagari’s calibration it does (`numerics/borrow.m`), and it does so almost linearly:

Borrowing limit	0	0.5	1	2	3	4	5	6
Threshold μ	3.42	3.50	3.58	3.74	3.89	4.05	4.21	4.38
Share of the gap closed	0%	11%	21%	40%	57%	74%	90%	104%

The natural limit here is $y_{\min}/r = 6.75$, and by a limit of 6 the gap is gone: the risky household’s threshold, 4.38, is the certainty-equivalent household’s, 4.33. So at this calibration the borrowing constraint accounts for the whole of the correction, and precaution accounts for none of it. Precaution is certainly present — at the loose limit the risky household still saves 1.62 against the certainty-equivalent 1.46, a precautionary excess of eleven per cent — but it moves the level of saving without moving the sign of its response to the interest rate. Across the other calibrations we ran (`numerics/borrow3.m`) loosening the limit always closes the gap and sometimes overshoots it, so the constraint is the dominant channel rather than an exact decomposition: with ample borrowing, risk can make Theorem 1 easier than certainty equivalence rather than harder.

For the programme this is the useful part. The missing ingredient for $\mu = 3$ is not a precautionary bound but a bound on what the borrowing constraint does to the comparative static, and that is a more tractable object: without a binding constraint the household’s Euler equation holds with equality at every state and Theorem 1 at zero wealth reduces to the duration criterion, which is proved outright for $\mu \leq 1$ and is an explicit finite computation in the primitives otherwise. What remains is the constrained states, which at this calibration are about eight per cent of the state space at every age.

5.8 The constraint channel is monotone, and needs no condition

The comparative static in the borrowing limit itself is a theorem, and unlike the one in the interest rate it needs nothing beyond the standing assumptions.

Theorem 5.3 (A tighter limit means less consumption and more saving). *Two households share every primitive and differ only in the borrowing limit, $f_2 \leq f_1$. Then at every age and every state in the tighter household’s region it consumes less and saves more.*

Formal statement: `IncomeFluctuation.finiteConsumption_le_of_floor_le`,
`IncomeFluctuation.le_finitePolicy_of_floor_le`.

The induction is the one of Theorem 5.1 with the interest rate held fixed, and that is exactly what removes the difficulty. Suppose the claim holds with k periods to come and fails at $k + 1$, so the tighter household consumes strictly more and therefore saves strictly less. Its Euler inequality under the cap, the looser household's Euler inequality above its floor — available because the looser household saves strictly more than the tighter one and hence strictly more than either floor — the induction hypothesis, and monotonicity of consumption in wealth chain into $u'(c_1) \geq u'(c_2)$, contradicting $c_1 > c_2$. The gross return appears on both sides and cancels, so there is no substitution term to control and no analogue of (1) is needed. Tightening the limit raises the marginal value of saving unambiguously, and the terminal stage is again free: a household with a lower floor runs down further in its last period.

That is the one-sided half of what the numbers showed, and it says the constraint channel is better behaved than the rate channel rather than merely different. What it does not yet give is a *size*: Theorem 5.3 orders the two consumption functions without bounding the distance between them, and it is the distance that turns into the 0.9 units of risk aversion between the constrained and certainty-equivalent thresholds. The natural next object is therefore a modulus for Theorem 5.3, and it has a clean form.

Theorem 5.4 (The modulus: a relaxation of Δ is worth at most Δ of wealth). *Let $\Delta = f_1 - f_2 \geq 0$ and $r \geq 0$. Then at every age and every state,*

$$c_k^{f_1}(x) \leq c_k^{f_2}(x) \leq c_k^{f_1}(x + \Delta).$$

Formal statement: *IncomeFluctuation.finiteConsumption_le_of_floor_le, IncomeFluctuation.finiteConsumption_le_shift.*

The household that may borrow down to f_2 therefore behaves, at wealth x , no more freely than the household confined to f_1 would at wealth $x + \Delta$: the entire value of the extra borrowing capacity is bounded by the capacity itself. The proof is the same backward induction, carried out at shifted arguments, and the shift pays for itself twice. The terminal stage holds with slack, since the looser household gains Δ by running down further while the shifted comparison gives it $(1 + r)\Delta$; and the interiority the step needs comes free, because the failure hypothesis forces the tighter household's saving to exceed the looser one's by more than Δ , hence to exceed $f_2 + \Delta = f_1$.

Two consequences. Since saving never falls with wealth, the sandwich gives the plain bound $0 \leq c_k^{f_2}(x) - c_k^{f_1}(x) \leq (1 + r)\Delta$, so the effect of the limit on consumption is at most the relaxation, up to one period's interest. And because the bound is stated as a shift in wealth rather than as an additive constant, it composes with the minimal propensity to consume: whatever upper bound on the propensity to consume out of wealth one has at age k , the modulus turns it directly into a bound on the consumption gap, and it is that product which becomes the gap between the constrained and certainty-equivalent thresholds.

5.9 The upper bound on the propensity to consume, and what it is worth

The missing piece is an upper bound on the propensity to consume out of wealth. The repository has the minimal propensity, which is a lower bound; the upper one follows by the same two-point Euler comparison, with no derivative taken. Write the bound as a secant slope in assets, so that s_k bounds $c_k(a_2) - c_k(a_1)$ by $s_k(a_2 - a_1)$.

Theorem 5.5 (The propensity to consume falls with the horizon). *In the last period the slope is exactly R (`stageConsumption_sub_zero`). If at stage k the slope is at most s and consumption is at least $c_{\min}$, then at a state where the poorer household saves, the slope at stage $k + 1$ is at most*

$$R \cdot \frac{\nu}{1 + \nu}, \quad \nu = \frac{cs}{c_{\min}},$$

with c the poorer household's consumption.

Formal statement: `IncomeFluctuation.stageConsumption_sub_le_of_slope`.

The step is the two Euler inequalities at the two wealth levels: the induction hypothesis bounds the rise in tomorrow's consumption by $s\delta$ with δ the difference in saving, which bounds the fall in marginal utility by the factor $(1 + s\delta/c_{\min})^{-\mu}$, and the budget identity $c_2 - c_1 = R(a_2 - a_1) - \delta$ turns that into the stated slope. Since $\nu \mapsto \nu/(1 + \nu)$ is a contraction towards $1 - 1/\nu$, the slope falls from R towards $R(1 - c_{\min}/(cs))$ as the horizon lengthens: the longer the life, the less of an extra unit of wealth is spent at once, which is the qualitative fact one wants.

It is not sharp enough. At Aiyagari's calibration, read at zero wealth in the highest earnings state, consumption is about 1.32 and the worst consumption next period about 0.88, so $\nu \approx 1.5s$ and the recursion settles at a propensity of about 0.36, against a true propensity there of about 0.04. Fed through Theorem 5.4 it bounds the effect of the borrowing limit on consumption by about 2.5 where the truth is about 0.04, a factor of sixty. The loss is the ratio of consumption today to the worst consumption tomorrow, and that ratio is where the sandwich's slack of order human wealth enters — the same obstruction that closed the infinite-horizon routes in the companion write-up, met here for the fourth time and in the same form.

So the chain now has every link proved except its sharpness. Theorem 5.2 gives the criterion without risk, Theorem 5.3 signs the constraint's effect, Theorem 5.4 bounds it by a shift in wealth, and Theorem 5.5 converts a shift in wealth into consumption. What none of them supplies is a bound on consumption that is tight to a few per cent rather than to a factor of ten, and every route we have tried reduces to exactly that.

5.10 Carrying earnings across all remaining ages

The two halves of the sandwich are not equally to blame. The upper half, $c_k \leq \kappa_k(m + H_k)$, carries the expected present value of future earnings and is about twenty per cent above the truth at Aiyagari's calibration. The lower half, $\kappa_k m \leq c_k$, ignores future earnings entirely and is a factor of ten below it. The repair is the same device applied on the other side: carry earnings forward across all the remaining ages instead of one step at a time.

Theorem 5.6 (The lower half, with human wealth). *Let H_k be the present value of k periods of the minimum earnings, $H_0 = 0$ and $RH_{k+1} = y_{\min} + H_k$. Then $\lambda_k(m + H_k) \leq c_k(m)$ at every age and state, where $\lambda_0 = 1$ and $\lambda_{k+1} = \beta\lambda_k R / (\beta + \lambda_k R)$.*

Formal statement: *IncomeFluctuation.minHumanWealth_le_stageConsumption.*

The old bound is the case $H = 0$. The propensity λ_k is the finite-horizon propensity of Theorem 2.1 discounted once more by patience at each step, and the extra discount is not decoration: without it the bound is false at the borrowing constraint, where the household consumes exactly its cash on hand while $\kappa_k(m + H_k)$ exceeds it, because H_k discounts earnings at R and $1/\kappa_k$ discounts at R/β . Discounting H_k at R/β instead repairs the corner and breaks the induction; the extra patience factor repairs both at once (`lowMPC_mul_minHumanWealth_1e`), and at $\beta = 0.96$, $r = 4\%$ and $\mu = 3$ it costs about three per cent over sixty periods.

At the state where the life cycle binds the household enters the next period with cash on hand about 2.0 and consumes about 0.88. The old bound gives 0.086; this one gives 0.33. Through Theorem 5.5 that improves the bound on the propensity to consume from about 0.94 to about 0.75, against a true propensity of 0.04.

So the multi-age argument buys a factor of four and not a factor of ten, and the reason is visible in the statement: it carries the *minimum* earnings, and a household that drew the minimum forever really would consume this little. What the programme needs is a lower bound carrying the *expected* human wealth, and that is not available from any chain of Euler inequalities, because the Euler inequality at a constrained state is exactly the statement that expected future earnings cannot be borrowed against. That is the one quantitative fact about a single household on which the whole programme turns, and identifying it precisely is what makes it attackable.

5.11 The constraint-incidence bound

Both halves can be carried at once. Accumulate human wealth by the *risk-adjusted* expectation — the power mean with exponent $-\mu$, which is what the Euler equation itself produces — and then *cap* it state by state at the level the borrowing constraint permits:

$$H_0 = 0, \quad H_{k+1}(z) = \min \left\{ \frac{1}{R} M_{-\mu}^z(y_w + H_k(w)), \left(\frac{1}{\kappa_{k+1}} - 1 \right) y_z \right\}$$

(`riskHumanWealth`), with κ the propensity of Theorem 2.1.

Theorem 5.7 (The constraint-incidence bound). *$\kappa_k(m + H_k(z)) \leq c_k(a, z)$ at every age and state.*

Formal statement: *IncomeFluctuation.riskHumanWealth_le_stageConsumption.*

The two ingredients meet exactly where they should. The power mean is what the Euler inequality delivers, and the step needs only that it is superadditive under a shift, which follows from its concavity and homogeneity (`powerMean_add_const_1e`). The cap is the constraint's incidence: at a corner the household consumes its cash on hand, which is at least y_z , and the cap is precisely the largest human wealth the bound may claim without exceeding that.

No extra patience factor is needed, because the cap does the work the patience factor did in Theorem 5.6.

At Aiyagari’s calibration, in the first period of a sixty-period life (`numerics/incidence.m`), the cap binds at the two lowest earnings states and nowhere else, which is where a household at zero assets is against the limit. The bound reaches

At zero wealth	lowest earnings	middle	highest
Bound as a fraction of the truth	1.000	0.890	0.877
Minimum-earnings bound, for comparison	1.010	0.409	0.295

and over the whole grid it never exceeds the truth, touching it exactly at the constrained corner. So the consumption bound is now within about twelve per cent where the life cycle binds, against a factor of three before. That is the sharpest bound in the development, and it was reached by bounding where the constraint binds rather than by another Euler chain.

It does not improve the propensity bound of Theorem 5.5, because that bound is driven by the worst earnings state next period, which is exactly where the cap binds. Closing the threshold argument would need the same sharpening on the worst state, and that turns out to be impossible within this class.

5.12 The worst state is pinned

Two facts settle it. The first is that the cap is not conservatism but necessity.

Theorem 5.8 (The cap is necessary). *Wherever a household at zero assets is against the borrowing limit, every affine lower bound $\kappa_k(m+H)$ valid at that state obeys $H \leq (1/\kappa_k - 1)y_z$.* Formal statement: `IncomeFluctuation.cap_necessary_of_corner`.

The proof is one line — at the corner consumption is exactly y_z — and the content is that Theorem 5.7 claims the largest human wealth any bound of its shape may claim. It cannot be sharpened within the affine class at all.

The second is that dropping the cap does not merely weaken the proof, it makes the bound false. Measuring effective human wealth as $c/\kappa - m$ at next-period assets of 1.67 (`numerics/worst.m`):

Effective human wealth at	truth	capped mean	uncapped mean	expectation
lowest earnings	10.96	5.99	13.11	16.57
highest earnings	27.92	23.90	26.38	32.24

At the lowest earnings state the uncapped power mean, 13.11, *exceeds* the truth, 10.96, so a bound built on it is invalid: the household will be against the limit later, and the uncapped recursion cannot see that. The truth sits between the two, and the distance is the price of capping at zero assets when the household in fact holds some. A bound that closed it would have to let human wealth depend on the assets held, which requires a lower bound on the household’s own asset path; the one available, from the upper sandwich, is negative at the state that matters.

So the worst state is pinned from both sides: the cap cannot be raised, because the corner forbids it, and it cannot be dropped, because the truth is below what dropping it would claim. That closes this family of arguments, and with it, for now, the route from a consumption bound to the threshold. What the programme has instead is an exact statement of what a sharper argument must supply: a lower bound on consumption that is asset-dependent rather than affine, and the asset path to support it.

5.13 The bound with assets in it

The obstruction just described is no longer binding, because the path bound it asked for is available. Proposition 4.2 read at the household's own earnings state rather than at the worst one is positive exactly where the cap bites hardest: a household in a high earnings state saves, so a period later it is not at the corner, and the cap that then applies to it is the one for a household with assets.

So assets go through the recursion twice over. Writing $\Phi_k(z, a)$ for that saving floor, clamped at zero,

$$G_0(z, a) = 0, \quad G_{k+1}(z, a) = \min\left(\frac{1}{R}M_{-\mu}^z(y_w + G_k(w, \Phi_{k+1}(z, a))), \left(\frac{1}{\kappa_{k+1}} - 1\right)(y_z + Ra)\right),$$

and the bound is $\kappa_k(m + G_k(z, a)) \leq c_k(a, z)$. The cap now grows with assets, and tomorrow's human wealth is read at the assets the household will hold rather than at zero.

Theorem 5.9 (The asset-dependent incidence bound). $\kappa_k(m + G_k(z, a)) \leq c_k(a, z)$, and $G_k(z, a) \geq H_k(z)$ for $a \geq 0$, so the bound is never worse than Theorem 5.7.

Formal statement: *IncomeFluctuation.assetHumanWealth_le_stageConsumption.*

, IncomeFluctuation.riskHumanWealth_le_assetHumanWealth

Measured against the truth over the whole grid at three ages (`numerics/assetdep.m`, `numerics/worstgrid.m`):

Worst ratio of bound to truth	$r = -7\%$	$r = 4\%$	$r = 10\%$
Theorem 5.7, cap at zero assets	0.237	0.606	0.760
Theorem 5.9, cap at the assets held	0.350	0.746	0.857

The new bound never exceeds the truth anywhere on the grid. The gain is largest exactly where the old one was weakest, at positive assets in low earnings states: at $r = 4\%$ with five units of assets and the lowest earnings, the ratio rises from 0.621 to 0.835. Where the old bound was already good, at zero assets, little changes, which is as it should be, since that is the state the cap was computed for.

What the refinement does not improve is the ceiling on aggregate capital, because the state that binds there is the highest earnings one, where the cap was never active: the implied ceiling at $r = -7\%$ is 22.15 against 22.17. That is still far below the 39.66 that feasibility alone gives, which is worth carrying into the existence argument of Section 4.

6 Theorem 1 at logarithmic utility

The slack condition of Section 4 is an Euler condition, and it is worth asking whether it settles the one case the stationary model does settle, $\mu = 1$. It does not, and the reason is instructive. Checking it on the whole grid at $\gamma = 1$ (`numerics/slacklog.m`), it holds at $r = 0$ and $r = 2\%$ and fails at $r = 4\%$ and above, by about twelve per cent in elasticity terms, at the youngest age with zero assets in the highest earnings state. Since a life-cycle equilibrium sits above the rate of time preference, 4.17% here, the Euler reduction says nothing where it is wanted, even at logarithmic utility.

6.1 A condition on the value gap

A different sufficient condition does hold. Write $V_k(\cdot; R)$ for the stage- k value.

Definition 6.1 (Value gap). The value gap has slope at least θ at stage k if, for $x \leq y$ in $[0, \bar{a}]$,

$$\theta(y - x) \leq [V_k(y; R_2) - V_k(y; R_1)] - [V_k(x; R_2) - V_k(x; R_1)].$$

Formal statement: `IncomeFluctuation.ValueGap`.

By the envelope theorem the slope of the gap is $R_2 u'(c_2) - R_1 u'(c_1)$, so the condition says the marginal value of assets rises with the rate. At $\gamma = 1$ that reads $c(R_2)/R_2 \leq c(R_1)/R_1$: consumption rises by proportionally less than the rate does.

Theorem 6.2 (Theorem 1 from the value gap). *If the value gap has a positive slope at stage k , then saving at stage k rises with the rate, at every state. Aggregate capital rises with it.*

Formal statement: `IncomeFluctuation.stagePolicy_mono_of_valueGap`.

`, IncomeFluctuation.olgCapital_le_of_valueGap`

Unlike the Euler route this needs no induction over ages. The condition on the continuation gives the conclusion for the household that faces it, directly, and the argument is Topkis with two continuations: the two households differ only in their cash on hand, which is ordered by the rate, and in their continuations, whose difference is increasing, so the higher-rate household cannot save less. Only concavity of period utility is used, as in Section 3.

6.2 Why it holds at $\mu = 1$, and only near there

At $\gamma = 1$ the patience factor is $\mathfrak{p} = \beta R$, so $q = \mathfrak{p}/R = \beta$ and the propensity $\kappa_k = 1/\sum_{i \leq k} \beta^i$ of Proposition 4.13 does not depend on the interest rate at all. Human wealth does, and falls as the rate rises. Where the borrowing constraint is slack, $c = \kappa_k(\bar{y} + Ra + H)$ and

$$R_2 c(R_1) - R_1 c(R_2) = \kappa_k [(R_2 - R_1)\bar{y} + R_2 H_1 - R_1 H_2] > 0,$$

which is the condition, with room. At a corner, $c = \bar{y} + Ra$ and the same inequality reduces to $R_1 \bar{y} \leq R_2 \bar{y}$.

Numerically (`numerics/condB.m`) the condition holds at $\gamma = 1$ at every age, state and rate tested, and fails from $\gamma = 2$, where between 19% and 51% of states violate it. It is

therefore sufficient and not necessary: Theorem 1 itself still holds at $\gamma = 2$, and up to about 3.4. So this route covers $\mu = 1$ and closes nothing above it.

What is proved, then, is the reduction, and it is a reduction to a condition that is true at logarithmic utility rather than one that is false there. Discharging the condition itself from primitives is not available from the sandwich of Theorem 2.1: the sandwich's width is $\kappa_k H_k$, and the rate response of exactly that quantity is what drives the gap, so the bounds cannot see it. That is the same obstruction the consumption-bound family ran into above, and it is the piece a sharper argument would have to supply.

7 The programme

The stages, in the order we intend to take them. The first is done (Section 2).

Equilibrium. A stationary overlapping-generations equilibrium: the distribution μ_1 of the youngest cohort is given (zero wealth, earnings drawn from the invariant law), μ_{j+1} is the pushforward of μ_j under the age- j policy and the earnings chain, capital supply is $K(r) = \sum_j \omega_j \int a d\mu_j$ with cohort weights ω_j , labour is exogenous, and the firm is Aiyagari's. Targets: continuity of K in the rate, a finite composition of continuous pushforwards; existence by the intermediate value theorem against the firm's demand, with the floor at the top of the interval from explicit accumulation over the life rather than from an ergodic identity; Light's Theorem 2 by induction on age, an ordering of policies at every age giving first-order stochastic dominance of every μ_j and so of K ; and single crossing against a decreasing demand.

Theorem 1 by backward induction. The research core: saving rises with the rate at every age, for $\mu \in \{3, 5\}$. The reduction of the companion write-up applies stage by stage, Theorem 1 at stage $k + 1$ following from the slack elasticity condition on the stage- k consumption functions at the pivot; the transfer lemma propagates an affine bound on the response of consumption to the rate backward from the terminal condition $c_0 = m$, where the response is exactly $\Delta r \cdot a$; and the slack condition at each stage is a finite check on the constants so obtained. The obstruction of the infinite horizon, a step multiplier of at least $1/p$, does not bite in J steps. The first session is numerical: a life-cycle economy with $J = 60$, Aiyagari's earnings chain and $\mu \in \{3, 5\}$, checking the slack condition at every age and wealth and running the recursion for the constants. If it holds the argument is formalised; if it fails only at moderate ages and wealth we look for the age-dependent refinement; if it fails badly, log utility is the fallback, where Theorem 1 at each stage follows from the rescaling proof of the companion, which is already an induction on Bellman iterates.

Diamond (1965) and Samuelson. The two-period economy without risk: a unique steady state and monotone global convergence for logarithmic utility and Cobb–Douglas production, existence under Inada conditions and a multiplicity witness in general, dynamic inefficiency of a steady state with $r < n$ by an explicit Pareto-improving transfer, the golden rule, and

the autarkic and monetary steady states of the pure-exchange economy. These are standalone results and will be the classical sections of this write-up.

Extensions. An age profile of earnings, retirement with a pension and a government budget, survival risk with accidental bequests as a second fixed point, population growth, and Lasry–Lions with age as time, whose finite-horizon core is already formalised and whose monotonicity is again the elasticity condition.

References

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- DIAMOND, P. A. (1965): “National Debt in a Neoclassical Growth Model,” *American Economic Review*, 55, 1126–1150.
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A Index of formal results

All in the namespace `LeanEconomics.IncomeFluctuation`, directory `LeanEconomics/OLG`. Every result depends only on the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`) and none uses `sorry`.

Result	Lean name	Module
Stage value, policy, consumption	<code>stageValue</code> , <code>stagePolicy</code> , <code>stageConsumption</code>	OLG/LifeCycle
Last period: no saving	<code>stagePolicy_zero</code>	OLG/LifeCycle
Last period: consume cash on hand	<code>stageConsumption_zero</code>	OLG/LifeCycle
Positive consumption at every stage	<code>stageConsumption_pos</code>	OLG/LifeCycle
Saving and consumption rise with wealth	<code>stagePolicy_mono</code> , <code>stageConsumption_mono</code>	OLG/LifeCycle
Carroll–Kimball at every stage	<code>concaveOn_stageConsumption</code>	OLG/LifeCycle
Euler under the cap, by stage	<code>stage_euler_le</code>	OLG/LifeCycle
Euler above the floor, by stage	<code>stage_euler_ge</code>	OLG/LifeCycle
Finite-horizon propensities, human wealth	<code>stageMPC</code> , <code>stageHumanWealth</code>	OLG/LifeCycle
Propensities in $(0, 1]$, above κ	<code>stageMPC_pos</code> , <code>stageMPC_le_one</code> , <code>minMPC_le_stageMPC</code>	OLG/LifeCycle
Lower Euler step	<code>crra_stageMPC_step</code>	OLG/LifeCycle
Upper Euler step	<code>crra_stageUpper_step</code>	OLG/LifeCycle
Theorem 2.1, lower	<code>stageMPC_mul_le_stageConsumption</code>	OLG/LifeCycle
Theorem 2.1, upper	<code>stageConsumption_le</code> , <code>stageHumanWealth</code>	OLG/LifeCycle
One-step saving bound	<code>stagePolicy_le</code>	OLG/LifeCycle

Result	Lean name	Module
Retired household (lump-sum pension)	<code>withPension</code>	OLG/Retirement
Cash on hand rises with assets	<code>withPension_resources_strictMono</code>	OLG/Retirement
Human wealth is state-independent	<code>stageHumanWealth_ withPension_ const</code>	OLG/Retirement
Perpetuity bound $H_k \leq p/r$	<code>stageHumanWealth_withPension_le</code>	OLG/Retirement
Life-cycle value with retirement	<code>lifeValue</code>	OLG/Retirement
Saving, consumption rise with wealth	<code>lifePolicy_mono, lifeConsumption_ mono</code>	OLG/Retirement
Pension schedules	<code>lumpSum, tapered, cliff</code>	OLG/MeansTest
Cash on hand	<code>cash</code>	OLG/MeansTest
Theorem 3.1	<code>isOptimalSaving_mono</code>	OLG/MeansTest
Monotone in assets when cash is	<code>isOptimalSaving_mono_assets</code>	OLG/MeansTest
Gentle taper keeps cash rising	<code>cash_strictMono_tapered</code>	OLG/MeansTest
A cliff reverses cash on hand	<code>cash_lt_cash_of_cliff</code>	OLG/MeansTest
Theorem 3.2	<code>exists_consumption_decrease</code>	OLG/MeansTest
Theorem 3.3, cliff	<code>exists_saving_decrease_in_assets</code>	OLG/MeansTest
Theorem 3.4	<code>policyOf_mono_of_lt</code>	OLG/MeansTest
One age's expectation operator	<code>stageStep</code>	OLG/ Equilibrium
Monotone class preserved	<code>monoRegion_stageStep</code>	OLG/ Equilibrium
Ordered policies, ordered functions	<code>stageStep_le_stageStep</code>	OLG/ Equilibrium
Life-cycle assets, aggregate capital	<code>cohortAssets, olgCapital</code>	OLG/ Equilibrium
Theorem 4.1	<code>cohortAssets_le, olgCapital_le_of_ stagePolicy_le</code>	OLG/ Equilibrium
Proposition 4.2	<code>le_stagePolicy</code>	OLG/ Equilibrium
Theorem 4.5	<code>olgEquilibriumRate_unique</code>	OLG/ Equilibrium
Argmax at a concave continuation	<code>argmax_ eq_ singleton_ of_ concaveSlices</code>	OLG/Continuity
Slice commutes with the operator	<code>sliceAt_ bellman, sliceAt_ augStageValue</code>	OLG/Continuity
Augmented stage policy	<code>augStagePolicy, augStagePolicy_eq</code>	OLG/Continuity
Jointly continuous stage policy	<code>continuousOn_augStagePolicy</code>	OLG/Continuity
Cohort recursion in the augmented state	<code>augCohortAssets, augCohortAssets_ eq</code>	OLG/Continuity
Theorem 4.6	<code>continuousOn_olgSupply</code>	OLG/Continuity
Theorem 4.7	<code>exists_olgEquilibrium</code>	OLG/Continuity
Saving rises iff consumption responds little	<code>stagePolicy_le_iff_consumption_le</code>	OLG/ RateMonotone
Terminal stage, unconditional	<code>stagePolicy_mono_zero</code>	OLG/ RateMonotone
Condition (1)	<code>StageSlack</code>	OLG/ RateMonotone

Result	Lean name	Module
Theorem 5.1, step	<code>stagePolicy_mono_step</code>	OLG/ RateMonotone
Theorem 5.1, induction	<code>stagePolicy_mono_of_stageSlack</code>	OLG/ RateMonotone
Theorem 5.1, aggregate	<code>olgCapital_le_of_stageSlack</code>	OLG/ RateMonotone
Human wealth, price of the path	<code>humanWealth, pathPrice</code>	OLG/Certainty Equivalent
Closed form $c_0 = W/D$	<code>consumption_mul_pathPrice</code>	OLG/Certainty Equivalent
Saving rises iff the cross product falls	<code>consumption_antitone_iff</code>	OLG/Certainty Equivalent
Theorem 5.2	<code>consumption_antitone_of_le_one</code>	OLG/Certainty Equivalent
Same household, different limit	<code>withFloor</code>	OLG/Borrowing Limit
Stages at a general limit	<code>finiteValue, finitePolicy, finiteConsumption</code>	OLG/Borrowing Limit
Theorem 5.3, consumption	<code>finiteConsumption_le_of_floor_le</code>	OLG/Borrowing Limit
Theorem 5.3, saving	<code>le_finitePolicy_of_floor_le</code>	OLG/Borrowing Limit
Theorem 5.4	<code>finiteConsumption_le_shift</code>	OLG/Borrowing Limit
Terminal slope is R	<code>stageConsumption_sub_zero</code>	OLG/ MPCBound
Theorem 5.5	<code>stageConsumption_sub_le_of_slope</code>	OLG/ MPCBound
Guaranteed human wealth	<code>minHumanWealth, lowMPC</code>	OLG/Lower Sandwich
The corner is respected	<code>lowMPC_mul_minHumanWealth_le</code>	OLG/Lower Sandwich
Theorem 5.6	<code>minHumanWealth_ stageConsumption</code>	<code>le_</code> OLG/Lower Sandwich
Power mean under a shift	<code>powerMean_add_const_le</code>	OLG/Incidence
Risk-adjusted, capped human wealth	<code>riskHumanWealth</code>	OLG/Incidence
Theorem 5.7	<code>riskHumanWealth_ stageConsumption</code>	<code>le_</code> OLG/Incidence
Theorem 5.8	<code>cap_necessary_of_corner</code>	OLG/Incidence
Definition 4.3	<code>StageRegions, allRegions</code>	OLG/LifeCycle
Sandwich on a reachable family	<code>stageConsumption_ stageHumanWealth_on</code>	<code>le_</code> OLG/LifeCycle
Floor on a reachable family	<code>le_stagePolicy_on</code>	OLG/ Equilibrium
Crude reachable bound	<code>reach, reach_mono</code>	OLG/Reachable
Proposition 4.4	<code>reachRegions</code>	OLG/Reachable

Result	Lean name	Module
Sandwich along a cohort's path	stageConsumption_ stageHumanWealth_reach	le_ OLG/Reachable
Floor along a cohort's path	le_stagePolicy_reach	OLG/ Equilibrium
Floor at the household's own state	le_stagePolicy_state_on	OLG/ Equilibrium
Proposition 4.8	olgCapital_le_of_invariant	OLG/Existence
Affine coefficients	cohortSlope, cohortFloor	OLG/Existence
Proposition 4.10	cohortFloor_ add_ mul_ le_ cohortAssets, le_olgCapital	OLG/Existence
Mean human wealth recursion	sum_mul_stageHumanWealth_succ	OLG/Existence
Theorem 4.11	stageMPC_ mul_ sum_ stageHumanWealth_le	OLG/Existence
The aggregate floor climbs	sum_mul_cohortFloor_mono, sum_mul_ cohortFloor_nonneg	OLG/Existence
Theorem 4.12	exists_ olgEquilibrium_ of_ bounds, exists_ olgEquilibrium_rate	OLG/Existence
Proposition 4.13	stageMPC_mul_mpcSum, stageMPC_mul_ one_sub_pow	OLG/LifeCycle
The geometric sum is bounded	sub_mul_mpcSum_le_one	OLG/LifeCycle
Constant-coefficient slope	cohortSlope_mul_mpcSum_succ	OLG/Existence
Proposition 4.14	mul_geomSum_le_cohortSlope	OLG/Existence
Human wealth in closed form	sum_mul_stageHumanWealth_eq	OLG/Existence
The floor in closed form	geomFloor_le_sum_mul_cohortFloor	OLG/Existence
Theorem 4.15	exists_ olgEquilibrium_ of_ geomBounds	OLG/Existence
Definition 6.1	ValueGap	OLG/Rate Monotone
Theorem 6.2	stagePolicy_ mono_ of_ valueGap, stagePolicy_mono_of_valueGap_all	OLG/Rate Monotone
Capital from the value gap	olgCapital_le_of_valueGap	OLG/Rate Monotone
The guaranteed asset path	assetFloor, assetFloor_ le_ stagePolicy	OLG/Asset Incidence
Human wealth with assets in it	assetHumanWealth, assetHumanWealth_mono	OLG/Asset Incidence
Theorem 5.9	assetHumanWealth_ le_ stageConsumption	OLG/Asset Incidence
It dominates the asset-free bound	riskHumanWealth_ le_ assetHumanWealth	OLG/Asset Incidence
Proposition 4.9	cohortAssets_ le_ of_ incidence, olgCapital_le_of_incidence	OLG/Existence
Existence with the sharper ceiling	exists_ olgEquilibrium_ of_ sharpBounds	OLG/Existence
Proposition 4.16	cohortCeil, cohortAssets_ le_ cohortCeil	OLG/Existence
Ceiling on aggregate capital	olgCapital_le_cohortCeil	OLG/Existence

Result	Lean name	Module
Theorem 4.17	<code>exists_olgEquilibrium_affine</code>	OLG/Existence
Proposition 4.18	<code>cohortFloorAt, cohortFloorAt_le_</code> <code>cohortAssets</code>	OLG/Existence
Floor on capital, clamped	<code>le_olgCapital_at</code>	OLG/Existence
Theorem 4.19	<code>exists_olgEquilibrium_clamped</code>	OLG/Existence